

Excitonic Insulator and the Extended Falicov–Kimball Model Away from Half-Filling

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We consider an extended spinless Falicov–Kimball model at an arbitrary doping level, focusing on the range of parameter values where a uniform excitonic insulator is stabilised at half-filling. In the two-dimensional case, we compare the properties of possible uniform phases and construct the Hartree–Fock phase diagrams, which include sizeable phase separation regions. It is seen that the excitonic insulator can appear as a component phase in a mixed-phase state in a broad interval of doping levels. In addition, in a certain range of parameter values the *excitonic metal* (doped excitonic insulator) is identified as the lowest-energy uniform phase. We suggest that this phase, which is unstable with respect to phase separation, may be stabilised when the phase separation is suppressed by the long-range Coulomb interaction. Overall, we find that excitonic correlations can affect the behaviour of the system relatively far away from half-filling.

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I. INTRODUCTION

Owing to its physical relevance and relative simplicity, the Falicov–Kimball model (FKM) continues to attract much attention ever since its inception[1] some 55 years ago. Theoretical investigations to date (involving a variety of mean-field, numerical, and rigorous approaches) can be grouped in two broad categories: (i) Investigations of the model (with additional extensions – Extended Falicov–Kimball model, EFKM) at half-filling $n = 1$, when the number of carriers equals the number of lattice sites[2, 3]. One of the most prominent directions here is related to the excitonic insulator (EI) – a gapped phase of condensed electron-hole pairs[4], which is stabilised within the EFKM under certain conditions (see Refs. [5–7] and many others). (ii) Studies of *doped* FKM with $n \neq 1$ (see, *e.g.*, Refs. 8–11), and those of closely related asymmetric Hubbard model[12–14]. Here, the ubiquitous finding is a strong tendency toward phase separation/segregation[8, 12, 14], whereby, for example, heavy electrons tend to congregate together in a part of the system only[9, 13]. It should be emphasised that all these studies without exception were restricted to the values of parameters which do not allow for a stable uniform EI phase in the half-filled case. This, in turn, implies that the EI is altogether left out of any phase-separation scenario.

The objective of the present work is to begin filling this gap. While a complete study would involve more sophisticated approaches and would also allow for phases modulated by a non-zero wave vector, here we restrict ourselves to a Hartree–Fock treatment and include only spatially-uniform phases (along with phase separation between these). The key findings can be summarised

as follows: (i) Phase separation is expected to take place over a broad range of values of parameters. In many cases, it includes a half-filled EI as one of the coexisting phases. (ii) The mean-field equations allow for a uniform EI-type solution with $n \neq 1$ in a broad range of concentrations n around half-filling, termed “excitonic metal”. While it is thermodynamically unstable with regard to phase separation, it presumably can be stabilised once the long-range Coulomb repulsion is included in the model.

The overall conclusion is that excitonic physics, which in the context of FKM was discussed in the half-filled case only, actually affects the behaviour of the system over a relatively broad range of values of parameters, including the doping level.

We construct phase diagrams of doped EFKM at low temperature, and discuss possible implications of our findings.

II. THE MODEL AND MEAN-FIELD EQUATIONS

We consider extended Falicov – Kimball model (EFKM) with a Hamiltonian

$$\mathcal{H} = -\frac{t}{2} \sum_{\langle ij \rangle} (c_i^\dagger c_j + c_j^\dagger c_i) + E_d \sum_i d_i^\dagger d_i + U \sum_i c_i^\dagger d_i^\dagger d_i c_i - \frac{t'}{2} \sum_{\langle ij \rangle} (d_i^\dagger d_j + d_j^\dagger d_i). \quad (1)$$

Here, the fermion operators c_i and d_i refer to electrons in the broad and narrow bands (nearest-neighbour hopping parameters $t > |t'|$), which interact via on-site repulsion U . We choose our units in such a way that both t and the period of the (d -dimensional hypercubic) lattice are equal to unity. When the energy shift E_d of the narrow

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band is equal to zero, the Hamiltonian is identical to that of an asymmetric Hubbard model (whereby one assigns opposite spins to the broad- and narrow-band carriers); on the other hand, the case of “pure” Falicov–Kimball model (as opposed to an *extended* one) is obtained from Eq. (1) in the limit $t' \rightarrow 0$.

We perform Hartree–Fock decoupling in the Coulomb term,

$$c_i^\dagger d_i^\dagger d_i c_i \rightarrow c_i^\dagger c_i n_d + d_i^\dagger d_i n_c - n_c n_d - d_i^\dagger c_i \Delta - c_i^\dagger d_i \Delta^* + |\Delta|^2, \quad (2)$$

where n_c and n_d are carrier densities in the broad and narrow band, and $\Delta = \langle c_i^\dagger d_i \rangle$ is the off-diagonal (excitonic) average value; the latter is referred to as induced hybridisation. We note that in the half-filled case, Hartree–Fock approximation yields a remarkably good agreement with quantum Monte Carlo simulations[5, 6].

While spatially modulated mean field solutions of EFKM may (and do) arise, here we will be considering uniform solutions only, hence suppression of the site index of the quantities Δ and $n_{c,d}$ (as well as of the net density, $n = n_c + n_d$). Furthermore, we are primarily interested in the region of parameters of Eq. (1), where the uniform EI state (characterised by $\Delta \neq 0$) is stabilised at half-filling, $n = 1$. This implies[5–7, 15, 16] that both $|E_d|$ and $|t'|$ exceed certain critical values (with $t' < 0$ and the critical value for $|t'|$ being numerically small). The value of U must be moderate in comparison to the width of the (broad) band, $U \lesssim 2d$.

The self-consistent mean-field equations for the *excitonic* phase are derived in a standard way (*e.g.*, by diagonalising the decoupled Hamiltonian) and take form

$$\Delta = \frac{1}{N} \sum_{\vec{k}} \frac{U \Delta (n_{\vec{k}}^1 - n_{\vec{k}}^2)}{\sqrt{(\xi_{\vec{k}} + t' \epsilon_{\vec{k}})^2 + 4|U \Delta|^2}}, \quad (3)$$

$$n_c - n_d = \frac{1}{N} \sum_{\vec{k}} \frac{(\xi_{\vec{k}} + t' \epsilon_{\vec{k}}) (n_{\vec{k}}^1 - n_{\vec{k}}^2)}{\sqrt{(\xi_{\vec{k}} + t' \epsilon_{\vec{k}})^2 + 4|U \Delta|^2}}, \quad (4)$$

which is valid at all values of bandfilling $0 < n < 2$,

$$n = \frac{1}{N} \sum_{\vec{k}} (n_{\vec{k}}^1 + n_{\vec{k}}^2), \quad n_{\vec{k}}^{1,2} = \left(e^{\frac{\epsilon_{\vec{k}}^{1,2} - \mu}{T}} + 1 \right)^{-1}. \quad (5)$$

Here, N is the total number of sites in the lattice, μ and T are chemical potential and temperature,

$$\epsilon_{\vec{k}}^{1,2} = \frac{1}{2} [E_d + U n + (1 + t') \epsilon_{\vec{k}}] \mp \frac{1}{2} \sqrt{(\xi_{\vec{k}} + t' \epsilon_{\vec{k}})^2 + 4|U \Delta|^2} \quad (6)$$

are quasiparticle energies in the two new bands, $\epsilon_{\vec{k}} = -\cos k_x - \cos k_y (-\cos k_z)$ is the tight-binding dispersion in two (three) dimensions, and

$$\xi_{\vec{k}} \equiv \xi(\epsilon_{\vec{k}}) = E_d + U(n_c - n_d) - \epsilon_{\vec{k}}. \quad (7)$$

Generally, the net energy of the excitonic phase (“excitonic metal”, see Sec. III below) can be evaluated as [cf. Eq. (2)]

$$E_{EM} = \frac{1}{N} \sum_{\vec{k}} \left(\epsilon_{\vec{k}}^1 n_{\vec{k}}^1 + \epsilon_{\vec{k}}^2 n_{\vec{k}}^2 \right) + U (|\Delta|^2 - n_c n_d). \quad (8)$$

At half-filling and at $T \rightarrow 0$, the quantity $n_{\vec{k}}^2$ vanishes, while $n_{\vec{k}}^1 \equiv 1$ for all values of $\vec{k}$; this corresponds to an EI, and Eq. (8) then yields E_{EI} , the energy of EI. We shall assume [without loss of generality – see Eq. (3)] that the quantity Δ is real and positive.

All other uniform mean field solutions do not have excitonic correlations (*i.e.*, $\Delta = 0$) and fall into two categories.

First, there can be up to two *single-band* solutions, which are characterised by one partially-filled band, the other band being either completely filled (at $n > 1$) or empty (at $n < 1$). These are prevalent for larger values of U or $|E_d|$, or closer to the end points $n = 0, 2$.

At smaller U , one also finds *semimetal* solutions with two partially-filled Hartree bands, centred around $U n_d$ (the broad band) and $E_d + U n_c = E_d + U(n - n_d)$ (the narrow band). Introducing $\lambda_c = \mu - U n_d$, we then find for the energy difference $|t'| \lambda_d$ between μ and the centre of the narrow band,

$$|t'| \lambda_d = \lambda_c - E_d + U(2n_d - n). \quad (9)$$

At $T \rightarrow 0$, the number of electrons in each band is found as

$$n_d = \int_{-d}^{\lambda_d} \nu(\epsilon) d\epsilon, \quad n - n_d = \int_{-d}^{\lambda_c} \nu(\epsilon) d\epsilon, \quad (10)$$

where $\nu(\epsilon)$ is the tight-binding density of states in the broad band, and its argument is measured from the band centre. Solving these three equations for the quantities n_d and $\lambda_{c,d}$ yields up to three solutions, and we have to keep the one which is characterised by the lowest energy,

$$E = E_d n_d + U n_c n_d + \int_{-d}^{\lambda_c} \epsilon \nu(\epsilon) d\epsilon + |t'| \int_{-d}^{\lambda_d} \epsilon \nu(\epsilon) d\epsilon. \quad (11)$$

The rest of the paper is concerned with numerical analysis of these solutions and of the ensuing phase diagrams in the interaction range $|t'| \ll U/2d \lesssim 1$ (weak to moderately strong coupling). We shall be interested in the low-temperature limit, $T \rightarrow 0$ throughout. While we consider the two-dimensional (2D) case only (aside from the Appendix), it is expected that the mean-field analysis of a three-dimensional (3D) system would yield qualitatively similar results.

III. THE UNIFORM SOLUTIONS AND THEIR INSTABILITIES

We begin with the excitonic phase, which at half-filling, $n = 1$, corresponds to the well-known excitonic insulator (EI) state, characterised by a non-zero value of

the off-diagonal average Δ and a gap of the order of $U\Delta$ in the electron spectrum. Properties of EI phase and its stability have been extensively addressed in the literature[2, 3, 7]. Presently we find, that once the EI solution is present at $n = 1$, a similar solution to the mean field equations, Eqs. (3–4), persists also in a certain range of doping values around $n = 1$. The chemical potential then lies within the upper or lower hybridised band, away from the excitonic band gap, hence such a phase should be more properly called *excitonic metal* (EM). This term, which was used previously in the context of doped Mott insulators[17], is now more commonly applied to the case of doped and/or otherwise imperfect EI, with a non-zero density of states at the Fermi level[18–20].

Fig. 1 illustrates the typical behaviour of EM solution in the case of moderate (a) or weak (b) Coulomb repulsion U . Numerical calculations are somewhat tedious, in particular because even relatively small values of t' strongly affect the EM bandstructure and may, *e.g.*, shift the minimum of the upper hybridised band away from $\vec{k} = 0$. For better convergence, our computations here and in Figs. 4 and 7 below included a very small temperature smearing (corresponding to $T = 10^{-5}$) in the equations (3–4) for the EM phase away from half-filling; this does not affect the results, which therefore represent the $T \rightarrow 0$ case.

For $U = 2$, the excitonic metal solution is found within the doping range $0.93 < n < 1.38$ and with increasing n the value of n_d increases from zero to one. Accordingly, at the two endpoints the excitonic solution merges with the two different single-band phases. At these endpoints, the off-diagonal average Δ shows square-root features; elsewhere, it has a maximum at $n = 1$ and a smoother feature near $n = 1.3$, the latter reflecting a feature in the bandstructure. In the weak-interaction regime of $U = 0.5$, the excitonic metal solution arises for $0.91 < n < 1.12$, merging with the semimetal solution at the endpoints.

Importantly, everywhere away from half-filling the compressibility $\partial\mu/\partial n$ is negative (see Appendix), signalling an instability with respect to an inhomogeneity formation and ultimately to phase separation. The dependence $\mu(n)$ shows a jump at $n = 1$. This point, where the value of μ lies within the gap between the lower (filled) and the upper (empty) bands [see Eq. (6)], has to be considered separately.

In this situation, the value of μ at $T = 0$ can be defined only as the $T \rightarrow 0$ limit of chemical potential at a finite T . In the latter case, the value of μ is fixed owing to the smearing of the Fermi distribution, which in turn gives rise to a non-degenerate gas of holes (electrons) in the valence (conduction) band. Typical evolution of $\mu(n)$ with decreasing temperature[21] is shown in Fig. 2.

We observe that at a finite temperature, $\mu(n)$ has a minimum below the point $n = 1$ and a maximum above it. Numerical data show that when the chemical potential attains its maximal (minimal) value, it lies within the energy gap and the energy difference between μ and the bottom of the conduction band (the top of the valence

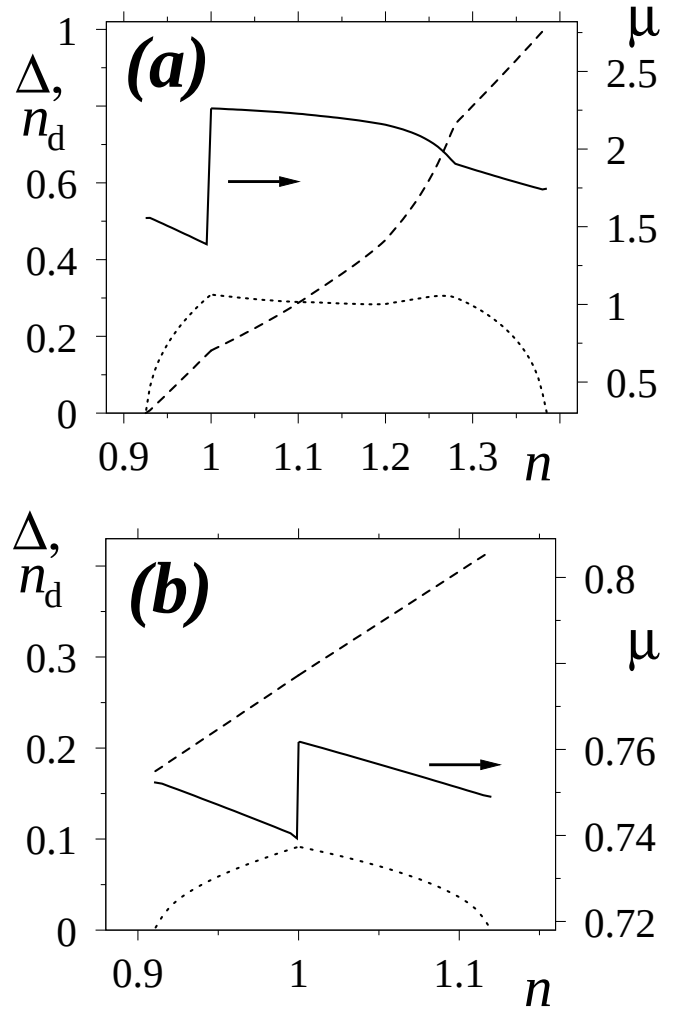


FIG. 1. Excitonic metal solution for $U = 2$, $E_d = 0.4$, $t' = -0.15$ (a) and for $U = 0.5$, $E_d = 0.4$, $t' = -0.015$ (b). Dashed and dotted lines show the dependence of n_d and Δ , respectively, on the carrier density n . Solid line (right scale) corresponds to the chemical potential μ .

band) is of the order of T . This entails two conclusions regarding the low-temperature limit.

First, the compressibility at $n = 1$ stays positive, increasing as the temperature decreases (and ultimately diverging at $T \rightarrow 0$). Thus, while excitonic metal at $n \neq 1$ is thermodynamically unstable, no such instability is found for an excitonic insulator at $n = 1$. This agrees with the literature, confirming the stability of EI in the suitable range of EFKM parameter values.

Second, as long as the chemical potential lies within the gap (and the distance from the gap edges is large in comparison with T), the compressibility is positive. At $T \rightarrow 0$, this translates into the following conclusion, which we will use in Sec. IV below: When the value of μ is externally fixed and lies anywhere within the gap, the EI phase (with $n \rightarrow 1$ at $T \rightarrow 0$) remains stable.

We now turn to the $\Delta = 0$ solutions mentioned in Sec. II above. The behaviour of $n_d(n)$ and $\mu(n)$ for single-

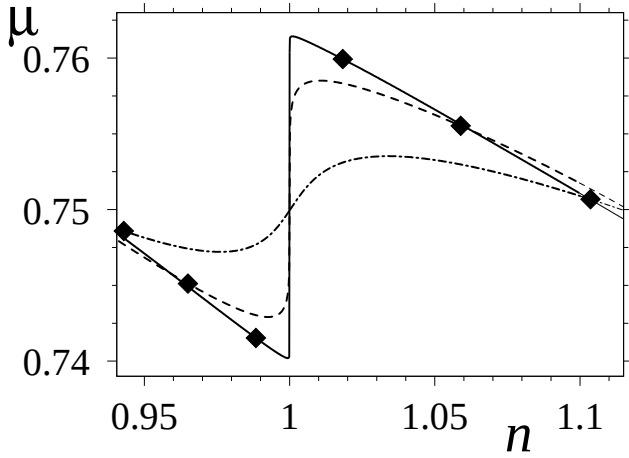


FIG. 2. Chemical potential near half filling in the excitonic metal phase for $U = 0.5$, $E_d = 0.4$, and $t' = -0.015$. Solid, dashed, and dashed-dotted lines corresponds to $T = 10^{-4}$, $T = 10^{-3}$, and $T = 3 \cdot 10^{-3}$ in the Fermi distributions in Eqs. (3–5). The diamonds correspond to respective chemical potential values crossing out of the hybridisation gap, whose width is about 0.02. The outer pair of diamonds refer to $T = 3 \cdot 10^{-3}$, and the middle one – to $T = 10^{-3}$.

band and semimetal solutions is illustrated in Fig. 3. We recall that when the value of U is sufficiently large, there can be up to three different semimetal solutions for a given value of n , and (at still larger U) up to two single-band solutions; we are always interested in the lowest-energy solutions of both types.

While in the weak coupling case (see Fig. 3 b) the sequential single-band and semimetal solutions evolve continuously from $n = 0$ to $n = 2$, in the larger- U case of Fig. 3 a we find a discontinuity. The latter reflects the presence of multiple semimetal solutions in the region around $n = 1.2$.

Single-band solutions, which are characterised by $n_d = 0$, $n_d = 1$, $n_d = n - 1$, or $n_d = n$, always have positive compressibility. This is not the case for the semimetal solutions, which in both cases shown in Fig. 3 ($U = 2$ and $U = 0.5$) have negative $\partial\mu/\partial n$ for certain values of n above half-filling. Indeed, the compressibility of semimetallic phase is given by

$$\frac{\partial\mu}{\partial n} = \frac{U^2\nu(\lambda_c)\nu(\lambda_d) - |t'|}{2U\nu(\lambda_c)\nu(\lambda_d) - \nu(\lambda_d) - |t'|\nu(\lambda_c)}, \quad (12)$$

[see Eq. (9)]; since $\nu(\epsilon)$ diverges at $\epsilon \rightarrow 0$ and equals $1/2\pi$ at the band edge, this can change sign at $U < \pi$ (in the case of small $|t'|$).

The energies of these single-band and semimetal solutions are plotted in Fig. 4. Importantly, whenever the excitonic metal solution is present, its energy $E_{EM}(n)$ [see Eq. (8)] is lower than that of other uniform solutions, as shown in the insets in Fig. 4. This is the typical situation, although a different behaviour can be found when all three of U , E_d , and n take larger values (see

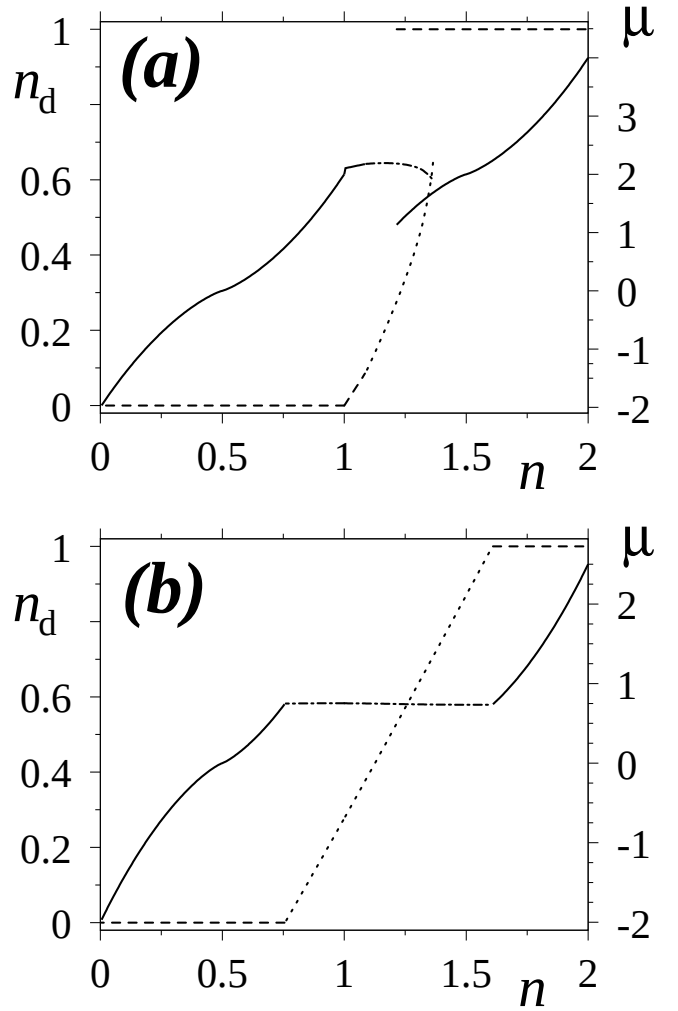


FIG. 3. Single-band and semimetal solutions for $U = 2$, $E_d = 0.4$, $t' = -0.15$ (a) and for $U = 0.5$, $E_d = 0.4$, $t' = -0.015$ (b). At a given value of carrier density n , dashed and dotted lines show the value of n_d (left scale) for the lowest-energy single-band and semimetal solutions, respectively. Solid and dashed dotted lines represent the corresponding values of chemical potential μ (right scale).

below, Sec. V). The peak at $n = 1$ (see insets in Fig. 4 a,b) is due to a sharp minimum of $E_{EM}(n)$ at half-filling.

Thus, the energy can be gained by opening the excitonic gap even away from half-filling, with the chemical potential lying within one of the hybridised bands. This is because unlike in a conventional low-energy, long-wavelength scenario, excitonic pairing in the EFKM involves short-range correlations, hence a restructuring of the spectrum over the entire bandwidth. The latter is illustrated in Fig. 5, where the dotted lines show the density of states (DOS) without hybridisation: broad conduction band and a delta-functional feature (bold dotted line) for the narrow band in the small- t' limit. Solid lines show the DOS in the two hybridised bands [see Eq. (6)]. These are shifted away from the original position of the narrow band, giving rise to a gap and also increasing the

overall bandwidth. The maxima of the DOS (usual logarithmic feature in 2D) are also moved away from the gap. The dashed lines denote the contribution of broad-band electrons to the DOS; we see that the localised electrons not only contribute the cusps adjacent to the gap, but also affect the DOS over the entire band. While the energy gain associated with opening the gap would be maximal at half-filling, in the case of Fig. 5 the carrier density n equals 1.2, and the chemical potential value for a hybridised solution is $\mu \approx 2.02$. Owing to the overall redistribution of the DOS, the energy of the Fermi sea is still lowered upon the gap opening.

We recall that while negative compressibility does indicate an instability with respect to phase separation, the former is far from being a necessary condition for the latter. Regardless of the sign of $\partial\mu/\partial n$ for single-band and semimetal solutions in the region where the lower-energy excitonic solution is present, the system always undergoes phase separation throughout this region (excepting the point $n = 1$). This is because the lowest-energy (excitonic) solution has negative compressibility. In Secs. IV, V we will see that this tendency towards phase separation actually extends well beyond the range of densities of the excitonic solution.

We remark that analysis of a “pure” Falicov – Kimball model [Eq. (1) with $t' = 0$] yields qualitatively similar behaviour of mean-field solutions, although quantitative changes associated with the non-zero t' are in some cases appreciable. We include a finite $t' < 0$ in our analysis due to the peculiar features of the $t' = 0$ case. These include an instability of excitonic insulator at $n = 1$ (for all values of U and E_d), as clarified in Refs. [6, 7]. For the numerical data shown in the plots, we chose the values of $-t'$ well above the respective critical values required to stabilise the homogeneous excitonic insulator state at half-filling; these critical values correspond to a second-order transition into a spatially modulated phase[6, 7].

It is well-known that at $n = 1$ the excitonic insulator also shows an instability of another kind, which arises in the immediate vicinity of the symmetric point $E_d = 0$ [for a broad range of values of t' in Eq. (1)]. There, previous work[5, 6, 15] reports a charge/orbital-ordered state, emerging via a first-order transition. Since presently we study uniform phases only, this ordering is beyond the scope of our approach, and therefore we do not detect any sign of the associated instability.

Finally, we note that in the Falicov–Kimball model the stability of excitonic insulator at $n = 1$ (away from the $E_d = 0$ point) can also be restored if, instead of the hopping t' in the narrow band, bare hybridisation is added to the model[7, 22]. This corresponds to replacing the last term in Eq. (1) with, *e.g.*,

$$V_0 \sum_i c_i^\dagger d_i - \frac{V_1}{2} \sum_{\langle ij \rangle} (c_i^\dagger d_j + c_j^\dagger d_i) + \text{H.c.}, \quad (13)$$

where the values of V_0 and/or V_1 exceed certain numerically small critical values. While this leads to numerical

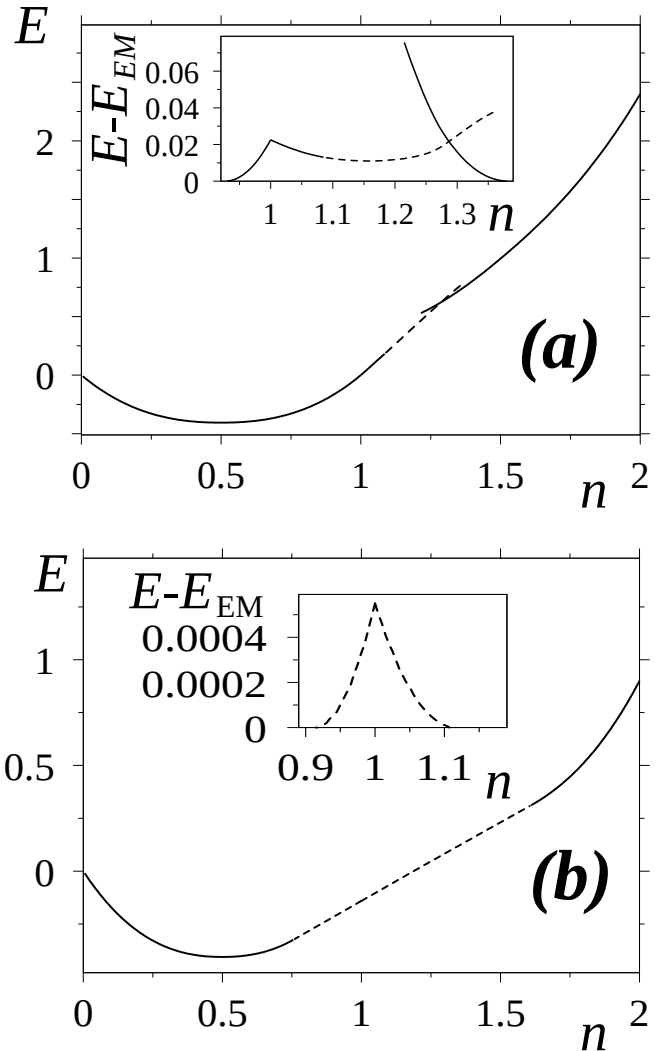


FIG. 4. Energies of single-band (solid line) and semimetal (dashed) solutions for $U = 2$, $E_d = 0.4$, $t' = -0.15$ (a) and for $U = 0.5$, $E_d = 0.4$, $t' = -0.015$ (b). Insets show the energy differences between these solutions and the excitonic metal, the latter always corresponding to the lowest energy.

changes, we tentatively verified that at the qualitative level, the conclusions of this section (that the excitonic metal state has a negative compressibility and also, typically, the lowest energy) remain valid. The important difference is that in the presence of a bare hybridisation one has to distinguish not between non-hybridised ($\Delta = 0$) and hybridised (excitonic metal or insulator) solutions as above, but rather between phases with small and large Δ . In the former case, Δ vanishes in the limit $V_{0,1} \rightarrow 0$, as the corresponding solution evolves into either single-band or semimetal one. This does not happen in the case of large- Δ solutions, corresponding to EM or EI (notwithstanding an instability at small $V_{0,1}$, which is similar to the instability at small t'). Throughout the rest of the paper, we will be including the effects of t' only.

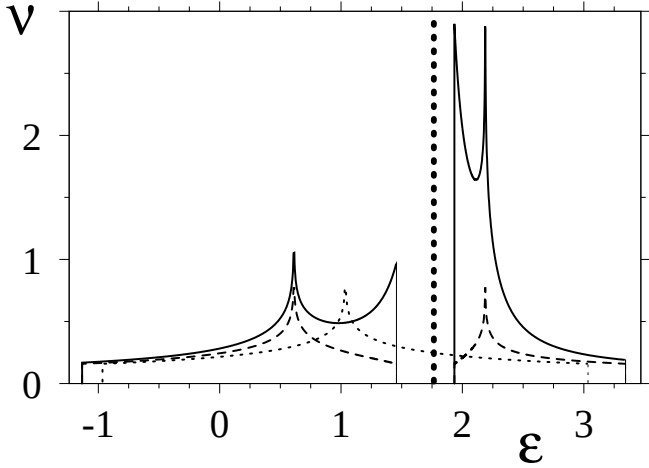


FIG. 5. Typical energy dependence of the quasiparticle density of states (DOS) for an excitonic metal solution (solid lines). Dashed lines show the contributions of broad-band electrons to the net DOS, whereas the dotted lines correspond to the DOS in the absence of hybridisation (obtained by formally setting $\Delta = 0$; bold dotted line represents the localised band). The data correspond to $n = 1.2$, $U = 2$, $E_d = 0.4$. We present results obtained for $t' \rightarrow 0$, as corrections due to a finite value of t' are not significant for our purposes here.

IV. PHASE SEPARATION

In the context of FKM and the asymmetric Hubbard model, the ubiquitous phenomenon of phase separation attracts much attention both in the half-filled[23, 24] and doped ($n \neq 1$, Refs. 9, 11–13) cases. However, the available studies of the doped case treat the situation away from the region of parameter values where the uniform EI state is stabilised for $n = 1$. As explained above, in terms of our Hamiltonian, Eq. (1), the latter region corresponds to finite non-zero values of *both* E_d and (negative) t' . In this case the EI, being the lowest-energy state for $n = 1$, can become one of the two component phases in a phase-separated system.

First, let us turn to a generic situation of equilibrium between two phases A and B, with respective energies $E_{A,B}(n)$ and chemical potentials $\mu_{A,B}(n)$, which depend on the density n . The equilibrium condition involves two equations for chemical potentials and Gibbs free energies:

$$\mu_A(n_A) = \mu_B(n_B), \quad (14)$$

$$E_A(n_A) - \mu_A(n_A)n_A = E_B(n_B) - \mu_B(n_B)n_B, \quad (15)$$

which determine the two values $n_{A,B}$ of density in the respective regions of phases A and B.

The preceding description refers to the situation when phases A and B are both gapless, and needs to be modified in the case of phase equilibrium between a gapless phase A and the (gapped) half-filled EI (we call this phase separation of the first kind, PS1). Chemical potential anywhere in the system will then be given by $\mu_A(n_A)$. As explained in the previous section, the EI will remain

stable (with $n_{EI} = 1$ at $T \rightarrow 0$) as long as μ lies between the top of the valence band and the bottom of the conduction band:

$$\epsilon_{max}^1 < \mu_A(n_A) < \epsilon_{min}^2 \quad (16)$$

[See Eq. (6)]. The value of n_A is then found from the equation for Gibbs free energies, which takes form

$$F_A(n_A) \equiv E_A(n_A) - E_{EI} - (n_A - 1)\mu_A(n_A) = 0 \quad (17)$$

[see Eq. (8)]. Conditions (16–17) take place of Eqs. (14–15). Since presently we consider uniform phases only, the phase A must be either single-band or semimetallic one. While this may leave out some possible scenarios, we do not expect this condition to be too restrictive: in principle, a (presumably gapped) homogeneous phase with spatial modulation may become stabilised near a commensurate value of density away from half-filling, yet this would require a large value of U . Even then, it might prove impossible to bring it in equilibrium with the half-filled EI.

We note that the question of EI stability at half-filling has been addressed by a variety of methods (see, *e.g.*, Refs. 5–7, and 15), also beyond the simple mean-field approximation. Within the range of parameter values where the uniform EI is stable at $n = 1$, this half-filled EI will therefore be stable also in a phase-separated state away from the (overall) half-filling, provided that the EI is in the thermodynamic equilibrium [see Eqs. (16–17)] with the other constituent phase.

There is an additional phase separation scenario (denoted PS2) which must be taken into account, in particular whenever the half-filled EI solution disappears at large $|E_d|$, and/or for at smaller U ($U \lesssim 0.75$). This corresponds to a phase separation into two different non-excitonic phases [A and B in Eqs. (14–15)], which can be either single-band with partially filled broad (SB1) or narrow (SB2) band, or semimetal. As usual, we can use Eq. (14) to express n_B as a function of n_A , and the phase equilibrium takes place at

$$F_{AB}(n_A) \equiv E_A(n_A) - E_B(n_B(n_A)) - [n_A - n_B(n_A)]\mu_A(n_A) = 0. \quad (18)$$

At smaller U , this can preempt the EI – single-band PS1 phase separation and may also render the EI state at $n = 1$ unstable (basically, the single-band state that would have been in thermodynamic equilibrium with the EI becomes unstable with respect to this second type of phase separation). We will now follow these two phase separation scenarios in a typical situation (the details may vary, depending on parameter values).

Let us imagine that we start with two fully occupied bands and gradually lower the concentration n . Provided that the value of E_d is less than half-bandwidth [more precisely, $E_d < 2 - 2|t'|$ in Eq. (1)], this leads to a depletion of carriers in the broad band and gives rise to a Fermi surface: the system is in a uniform SB1 phase, denoted phase A. As the value of n decreases further, the

value of the chemical potential μ_A crosses (from above) inside the range, corresponding to the spectral gap of a half-filled EI, Eq. (16), with the value of $F_A(E)$, Eq. (17), being negative. Also, as we lower the carrier density from the $n = 2$ endpoint, we eventually arrive at a point where a solution to Eq. (14) appears [see the two solid lines in Figs. 3 (a,b)], with phase B being another single-band phase (or possibly a semimetallic one). Initially, the value of F_{AB} , Eq. (18), will also be negative.

With lowering n further, the stability of the uniform phase A is lost at a point $n = n_A^*$ where either F_A or F_{AB} vanishes (whichever occurs first). In the case where $F_A(n_A^*) = 0$ [while either $F_{AB}(n_A^*) < 0$ or Eq. (14) cannot be solved], lowering n further leads to phase separation into phase A (with $n_A = n_A^*$) and the EI phase; the EI fraction expands until phase A disappears at half-filling ($n = 1$) and the system turns into a uniform EI. At this point, the value of the chemical potential suffers a negative jump (within the EI spectral gap), and reducing n further leads to a phase separation into EI and a single-band phase with empty narrow band.

If, on the other hand, $F_{AB}(n_A^*)$ vanishes [while $F_A(n_A^*)$ is still negative], this is followed by phase separation into phase A (with $n_A = n_A^*$) and phase B , with the density n_B^* determined by $\mu_B(n_B^*) = \mu_A(n_A^*)$. The fraction of phase B then increases until we reach the point $n = n_B^*$, beyond which the system remains in the uniform phase-B state with decreasing $n_B = n$. In the case where $n_B^* < 1/2 < n_A^*$, the EI state does not arise: at half-filling, its energy E_{EI} is larger than the energy of an appropriate A-B phase mixture,

$$[(1 - n_B^*)E_A(n_A^*) + (n_A^* - 1)E_B(n_B^*)] / (n_A^* - n_B^*). \quad (19)$$

These two situations are exemplified by Fig. 6. For the values of parameters used in Fig. 6 (a), Eq. (16) is satisfied at $n < 1.68$, and $F_A(n)$ changes sign at $n_A^* \approx 1.522$, signalling phase separation into a half-filled EI and the single-band phase A with $n = n_A^* > 1$ (phase separation of the type PS1). On the other hand, the value of F_{AB} , which would describe phase separation into two different single-band phases, changes sign at $n \approx 1.492 < n_A^*$, and this latter scenario is therefore irrelevant (as the uniform phase A at this doping level is already unstable with respect to the other type of phase separation). In the case of Fig. 6 (b), Eq. (14) for those parameters values can be solved only in a very narrow doping range $1.608 < n < 1.612$ [cf. Fig. 3 (b), where the overall negative slope of the dashed line in the centre of the figure is very small]. The value of $F_{AB}(n)$ changes sign at $n_A^* \approx 1.611$, whereas $F_A(n)$ vanishes at $n \approx 1.610 < n_A^*$ and the EI state is therefore irrelevant. Phase separation is thus of the PS2 type, corresponding to a mixture of two single band phases with $n = n_A^*$ and $n = n_B^* \approx 0.756$. We note that the precise boundary between the two scenarios depends also on the value of t' , as increasing the latter tends to tilt the balance in favour of PS1.

While generally the PS2 phase separation may also involve SB2 and semimetal phases, the specific case above

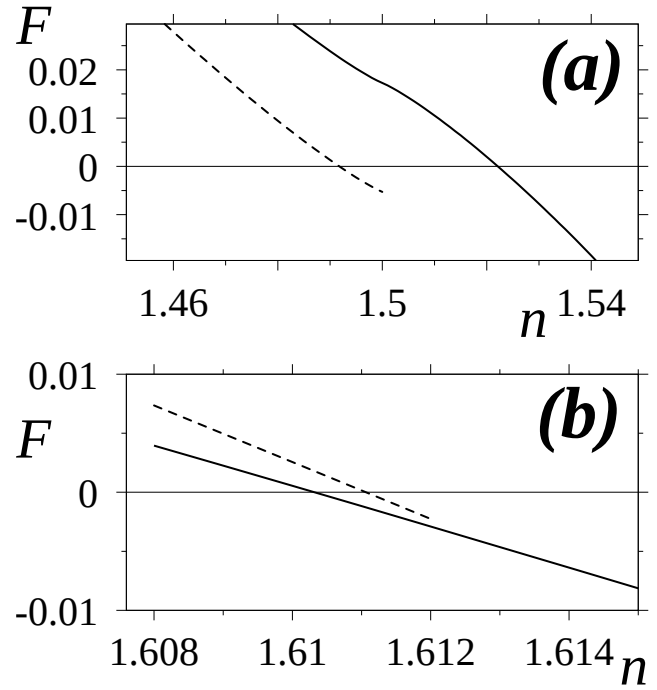


FIG. 6. Values of $F_A(n)$ (solid) and $F_{AB}(n)$ (dashed) for $U = 2$, $E_d = 0.4$, $t' = -0.15$ (a) and for $U = 0.5$, $E_d = 0.4$, $t' = -0.015$ (b).

corresponds to a phase equilibrium between two different SB1 phases (with filled and empty narrow band). Apparently, this can be identified as a situation found in numerical investigations of doped FKM and asymmetric Hubbard model, whereby the narrow-band electrons tend to clump together in a part of the system (see, *e.g.*, Refs. 11 and 13)

We are now finally in a position to discuss the phase diagrams emerging from our study.

V. THE PHASE DIAGRAMS

In Fig. 7 we present the phase diagrams corresponding to different values of U and t' . The values of the latter were chosen to be sufficiently large, so that the EI state at half-filling is expected to be stable with respect to low-lying collective excitations [7]. On the other hand, the bare bandwidth of the narrow band remains several times smaller than U in all cases. The phase diagrams are of course symmetric under the transformation $E_d \rightarrow -E_d$, $n \rightarrow 2 - n$.

Phase separation is confined to the shaded regions of the phase diagrams. The red region corresponds to PS1, whereby one of the two component phases is the half-filled EI. As explained in the previous section, the state of the system for any $n > 1$ ($n < 1$) corresponds to an appropriate mixture of the EI phase and the uniform phase which borders the PS1 region above (below) half-filling at the same E_d , with the value of density n at the

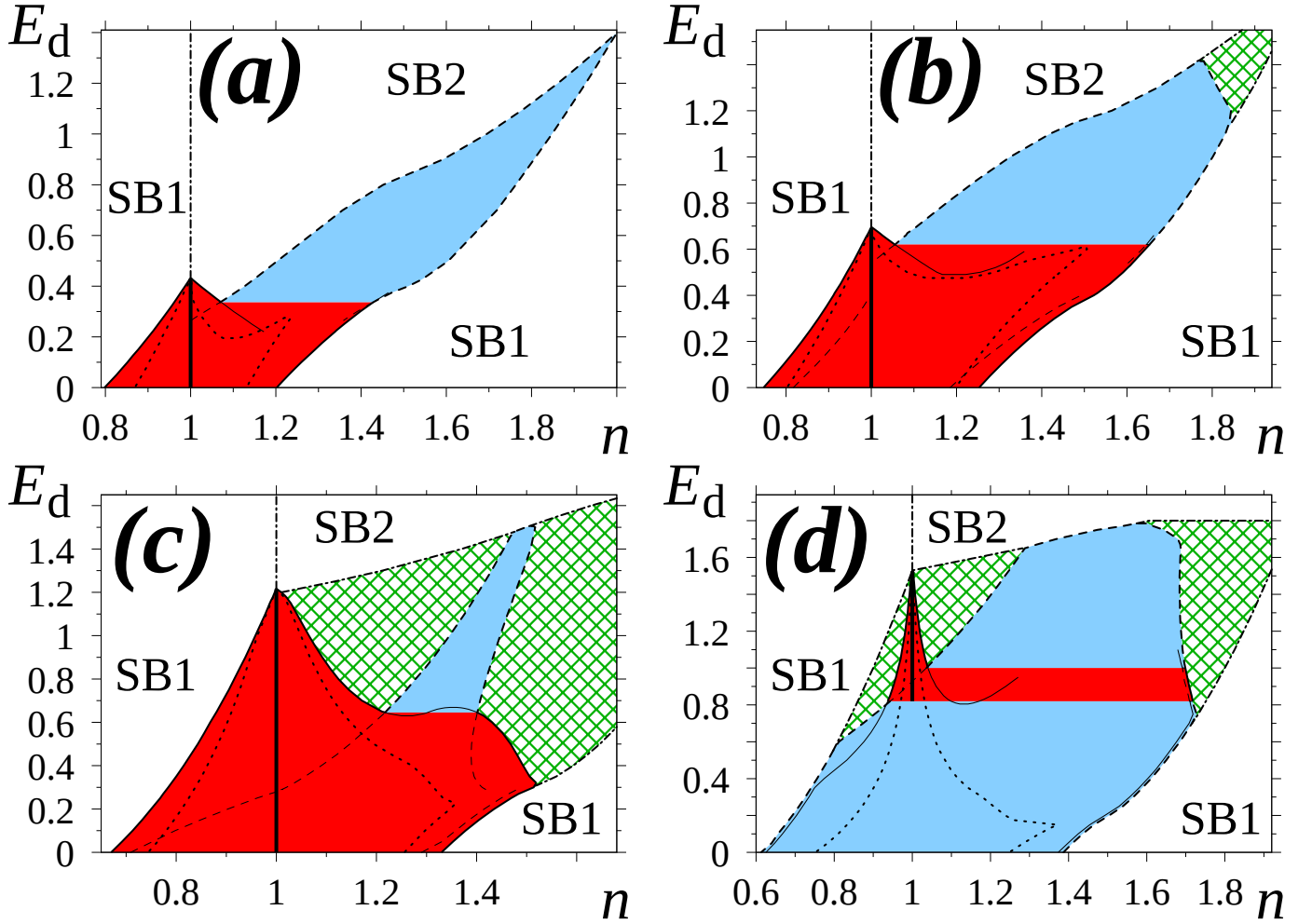


FIG. 7. (colour online) Calculated phase diagrams for the 2D EFKM with $U = 4$ and $t' = -0.3$ (a), $U = 2$ and $t' = -0.15$ (b), $U = 1$ and $t' = -0.1$ (c), $U = 0.5$ and $t' = -0.015$ (d). The shaded regions correspond to phase separation: red (darker grey in the black and white version) to PS1, whereby one of the component phases is the half-filled EI state, and light blue (lighter grey) to PS2. Solid and dashed lines correspond to solutions of Eqs. (16–17) and (14),(18) [PS1 and PS2 conditions, respectively]; bold vertical solid lines at $n = 1$ denote a stable single-phase EI. Dotted lines are the boundaries of the region around $n = 1$, where the EM solution exists and has the lowest energy among the uniform single-phase states. Outside the phase separation region, the dashed-dotted lines show boundaries between single-band phases SB1 and SB2 (with the Fermi level within the broad or the narrow band, respectively) and semimetallic state (hatched).

border. Likewise, the state of the system within the PS2 (light-blue) region is the mixture of the two phases which border the phase separation region from right and left at a given E_d .

The stable EI phase is represented by a thick vertical solid line at $n = 1$. The upper tip of this line corresponds to EI formation in a semiconductor, whereas in the lower part excitonic pairing occurs in a metallic “parent” state. In order to locate the crossover between these two regimes, we have to formally set the off-diagonal average Δ to zero and require that the conduction and valence Hartree bands touch:

$$Un_d + 2 = E_d + Un_c - 2|t'|. \quad (20)$$

While in the weakly-interacting case of Fig. 7 d this equality is satisfied (within our numerical accuracy) at

the upper point of the EI line, $E_d \approx 1.525$, in case of the strong interaction (Fig. 7 a) this takes place at $E_d \approx 0.24$, near the middle of the EI line which extends up to $E_d \approx 0.43$. In the intermediate cases of Fig. 7 b and c, Eq. (20) is satisfied, respectively, at $E_d \approx 0.56$ and $E_d \approx 1.205$, while the upper end of EI line corresponds to $E_d \approx 0.695$ and $E_d \approx 1.215$.

The observation that *the PS1 area (and associated excitonic behaviour) can extend relatively far away from the $n = 1$ line* is among the main results of the present study. We see that the PS1 region is most prominent at the intermediate values of U . At large U (Fig. 7 a) it is suppressed due to large Hartree contribution to particle energy in the EI state, whereas at small U (Fig. 7 d) the energy gain associated with the excitonic pairing

becomes marginal.

Bold solid (dashed) lines at the borders of the PS1 (PS2) regions correspond to solutions to Eqs. (16–17) [for PS2, Eqs. (14) and (18)]. Lighter lines correspond to continuations of these solutions within the phase separation regions; while these do not correspond to any transitions (see above, Sec. IV), they help visualise the overall structure of the phase diagram. We note that all four diagrams, which sweep different parameter regimes from strong to weak interaction, are remarkably similar in this regard. Therefore there is no doubt that results for other parameter values would be qualitatively similar.

The parameter space outside the PS1 and PS2 regions is occupied, for the most part, by the single-band phases SB1 (with partially-filled broad band) and SB2 (whereby the chemical potential lies within the narrow band). In Fig. 7, the boundaries of the corresponding regions are shown with dashed-dotted lines. While the boundary at half-filling is continuous[25] (with increasing density, the broad band is filled at $n = 1$ and subsequently the filling of the narrow band begins), no direct continuous transition between SB1 (with filled narrow band) and SB2 phases is possible in the $2 > n > 1$ region. In the large- U case (see Fig. 7 *a*), the intervening area is entirely taken over by phase separation, PS2. In a completely filled system at $n = 2$, the difference between SB1 and SB2 phases disappears, hence the boundaries of these two regions must meet (which is indeed the case for all parameter values used in Fig. 7). This occurs when the upper edges of the two filled Hartree bands coincide,

$$U + 2 = U + E_d + 2|t'|,$$

or $E_d = 2 - 2|t'|$. The other end of the SB2-region lower boundary (more precisely, its continuation within the PS1 region) crosses the EI line at half-filling in the general area of a crossover between semiconducting and metallic excitonic pairing [see Eq. (20)], as can be expected.

Another non-excitonic phase, a semimetal with two partially-filled bands, is strongly disfavoured at large U due to a large Hartree contribution to its energy. We see that as the value of U decreases, areas of semimetallic phase (hatched) emerge in the intervening region between SB1 and SB2. For smaller- U cases shown in Fig. 7 *c,d*, we see that at certain values of E_d the phase separation (PS2) may occur also between semimetallic phases with different values of density. This is because uniform semimetallic solutions typically possess negative compressibility in a certain range of values of n [see above, Eq. (12)]; in addition, with varying density a discontinuous switching between different semimetallic solutions (see Sec. II) can occur, also entailing phase separation.

Within our model, Eq. (1), the excitonic metal (EM) phase with $n \neq 1$ has a negative compressibility (see Sec. III and Appendix) and does not appear on the phase diagram. Dotted lines within the phase separation regions in Fig. 7 show the areas around half-filling where the EM phase has the lowest energy among the uniform single-

phase solutions. For weaker U of Fig. 7 *c,d* the latter is always the case whenever the EM solution exists. For $U = 2$, (see Fig. 7 *c*), there is a narrow region immediately above the upper (concave) boundary of the EM area at $n > 1$, where the EM solution is present, yet has a higher energy than a single-band one. Within and around this region we sometimes encounter a situation where several EM solutions are present; the lowest-energy one has a negative compressibility. Finally, at $U = 4$ this region extends upwards and to the right far into the larger- E_d, n range, although in some cases the EM solution there may be spurious (corresponding to a local energy *maximum*). Elsewhere, the boundary of the EM region corresponds to a continuous transition ($\Delta \rightarrow 0$) into the lowest-energy uniform non-excitonic phase. We also note that the EM area significantly expands whenever the absolute value of t' is decreased. We will continue our discussion of the EM phase in the following section.

We recall that our selection of single phase states and component phases for phase separation includes uniform mean-field solutions only. We expect that this affects the validity of our results primarily at commensurate fillings, especially at $n = 1$ beyond the region where the uniform EI phase is stable (which includes the charge/orbital ordering at small $|E_d|$, see Refs. 5, 6, and 15). Possible instability of a uniform semimetal at a fractional filling with respect to charge/orbital modulation is another issue which falls beyond the scope of this work. Finally, when the narrow-band hopping t' is decreased below the critical value required to stabilise the uniform EI, the EI state at $n = 1$ acquires spatial modulation of both charge density and excitonic correlations[6]. We expect that this should not significantly affect the behaviour in the doped regime at a small but finite t' : within the overall picture described above, the modulated EI would take place of the uniform EI as a phase component in the PS1 region; apart from a quantitative change, the overall structure of the phase diagram presumably remains unaffected.

VI. DISCUSSION

The possibility of excitonic condensation, resulting in a formation of EI state in a metallic or semiconducting compound at a low temperature, attracts broad experimental and theoretical effort (see Ref. 26 for a contemporary review). This includes theoretical studies of doped systems, with applications to hexaborides[18, 27–29], twisted bilayer graphene[19], and other compounds[20]. These broader-band systems are generally treated within the low-energy, long-wavelength approach, which in its original form involves re-structuring of the spectrum in the immediate vicinity of the (nested) Fermi surfaces, and lowering the net energy slightly by opening a narrow gap at the Fermi level[30]. When doped, the proximity of the EI state affects the properties of the system only as long as the chemical potential lies close to the (possibly smeared) excitonic gap, corresponding to a

narrow range of doping values (cf. Appendix). Within this range, a rich physical picture emerges once additional degrees of freedom (most notably spin) and features like imperfect nesting[18, 19, 27] or presence of impurities[28] are taken into account. Findings include, *inter alia*, ferromagnetism[18, 27–29, 31, 32], excitonic metal behaviour[18–20, 32], and phase separation[18, 27, 29].

However, to the best of our knowledge the available theoretical literature on the extended Falicov – Kimball model away from half-filling does not address the issue of excitonic correlations. It should be emphasised that EFKM corresponds to a rather different realisation of the EI, whereby the gap is comparable or larger than the width of the narrow band[33], and the issue of Fermi surface nesting is no longer relevant. The excitonic correlations in the EFKM are inherently short-wavelength, hence the entire spectrum throughout the Brillouin zone is modified. The EI gap itself is sufficiently broad to allow for an equilibrium with a conducting phase in a wide range of carrier densities [we recall that the chemical potential of this second phase, which depends on density, must be located within the EI gap, see Eq. (16).]

Indeed, in Sec. V above we saw that under the right conditions, whenever the EI state is stabilised at half-filling, it persists in a doped system as a component phase in a phase-separated state in a relatively broad range of carrier densities. In such a state, the electrical current would be carried by the other phase component only, and percolative transport behaviour is anticipated. Since the chemical potential lies within the EI energy gap, the *average* density of states will show a broad depression around the Fermi level. There is a number of intriguing physical issues, including the Andreev-like scattering of carriers by the borders of the EI regions[34] and the overall evolution of the system with increasing temperature (cf. Ref. 35).

There is also another possibility, which might turn out to be relevant for actual physical systems. These always include the long-range Coulomb interaction, which dictates that single-phase areas in a phase-separated system cannot grow beyond a certain size; both the Coulomb contribution and the surface tension of the interphase boundaries[36] are increasing the energy of the phase-separated state with respect to that of a homogeneous one. Since the energy difference between phase-separated state and competing single-phase states is typically rather small, this may destroy the phase separation/inhomogeneity and stabilise the homogeneous behaviour. On the other hand, the *relative* energies of various uniform single-phase solutions at a given value of carrier density remain unaffected.

In Sec. V we saw that there is a sizeable region on a phase diagram, where the excitonic metal phase has the lowest energy among the uniform homogeneous states (see the dotted lines in Fig. 7). The fact that *in the absence of the long range Coulomb interaction* the EM state was found to have a negative compressibility (Sec.

III) obviously has little bearing on the situation when the Coulomb interaction is present. We therefore suggest that including the Coulomb interaction might stabilise the homogeneous EM phase in a doped system.

Should this be the case, the density of states at the Fermi level will be *increased*, and a broad energy gap will open above or below the chemical potential. The (hybridised) carriers at the Fermi level will have predominantly narrow-band character (see Fig. 5), which will in turn affect the transport properties.

Either way, we expect that EI behaviour, which is conventionally associated with EFKM at half-filling, may affect the properties of a doped system over a broad range of carrier densities. Therefore, whenever experimental results suggest that the EI or EM behaviour persists beyond one or two percent doping, this may imply that the EFKM-like picture of strong short-range correlations is relevant. In particular, this might be the case for 1T-TiSe₂ (where the charge density wave, broadly attributed to excitonic pairing, is cut off by a superconducting state at 4%-6% copper intercalation[37]), and likely also for Ta₂NiSe₅ (see Refs. 38 and 39).

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Appendix A: Compressibility of the excitonic metal phase

At the phenomenological level, the origin of the negative compressibility in the EM phase away from half-filling is clear from Fig. 1. Both above and below the half-filling the EM eventually merges with another phase, and the difference between the values of μ at these two points, whether negative (as in Fig. 1 *b*) or positive (Fig. 1 *a*), is relatively small. However, the dependence of $\mu(n)$ in the intervening EM phase must accommodate a large *positive* jump at $n = 1$, equal to the width of the EI energy gap. It is therefore quite natural that away from this jump the value of μ decreases with n .

In order to analyse the underlying mechanism in more detail, we will turn to the $t' \rightarrow 0$ case, ignoring the other (collective-excitation) instability[7] which arises there. This is expedient because the negative compressibility of EM persists for $t' \rightarrow 0$ as well and clearly has the same nature as for $t' \neq 0$, while on the other hand the $t' = 0$ case is much more transparent analytically.

At $t' = 0$, the quasiparticle energies $\epsilon_{\vec{k}}^{1,2} = \epsilon_{1,2}(\epsilon_{\vec{k}})$, Eqs. (6–7), are monotonously increasing functions of the tight-binding energy $\epsilon_{\vec{k}}$. Thus for $n < 1$ ($n > 1$) the quasiparticle states in the lower (upper) band are occupied for $\epsilon_{\vec{k}} < \epsilon_0$, and the chemical potential μ is given

by $\epsilon_1(\epsilon_0)$ [$\epsilon_2(\epsilon_0)$ for $n > 1$]. When the carrier density is increased by a small δn , the chemical potential change $\delta\mu = \delta_r\mu + \delta_c\mu$ is obtained from Eq.(6) as a sum of the “rigid” bandstructure contribution,

$$\delta_r\mu = \frac{1}{2} \left(1 \pm \frac{\xi_0}{\sqrt{\xi_0^2 + 4U^2\Delta^2}} \right) \delta\epsilon_0 \quad (\text{A1})$$

[with $\delta\epsilon_0 = \delta n/\nu(\epsilon_0)$ and $\xi_0 = \xi(\epsilon_0)$, see Eq. (7)], which is always positive, and the interaction-induced “correlation” term,

$$\delta_c\mu = \frac{U\delta n}{2} \mp \frac{\xi_0 U(\delta n_c - \delta n_d) + 4U^2\Delta\delta\Delta}{2\sqrt{\xi_0^2 + 4U^2\Delta^2}}, \quad (\text{A2})$$

which includes the effects of a self-consistent change of $n_{c,d}$ and Δ . The upper and lower signs in Eqs. (A1–A2) correspond to $n < 1$ and $n > 1$ respectively.

By varying n in the mean field equations (3–4) and solving for $\delta n_c - \delta n_d$ and $\delta\Delta$, we find that $\delta_c\mu$ is proportional to δn ,

$$\delta_c\mu = \frac{(2\xi_0 Y_0 - 2U\Delta Y_1 - U\Delta)^2 + (4U\Delta Y_0 + \xi_0 Y_1)^2}{16Y_0^2 + 4Y_1^2 + 2Y_1} \times \frac{2U\delta n}{\xi_0^2 + 4U^2\Delta^2}, \quad (\text{A3})$$

where [cf. Ref. 7, Eqs.(B3-B4)]

$$Y_0 = \frac{1}{2} \int \frac{U^2 \xi \Delta \nu(\epsilon) d\epsilon}{(\xi^2 + 4U^2\Delta^2)^{3/2}}, \quad Y_1 = - \int \frac{2U^3 \Delta^2 \nu(\epsilon) d\epsilon}{(\xi^2 + 4U^2\Delta^2)^{3/2}}, \quad (\text{A4})$$

$\xi = \xi(\epsilon)$, the integration limits are from $-d$ to ϵ_0 for $n < 1$ or from ϵ_0 to d for $n > 1$, and $d = 2$ or 3 is the dimensionality.

The denominator in Eq. (A3) is *negative*, as proved in Ref. 7 (Appendix B). The proof, based on the required stability of the EI state at a fixed $n = 1$ with respect to an “external field” V_0 [see Eq. (13)], can be directly generalised for the case of an EM state away from half-filling.

The quantity $\delta_c\mu/\delta n$ is therefore negative, and gives a negative contribution to $\partial\mu/\partial n$. This has a clear physical

meaning: while in the absence of correlations the compressibility would have been given by $\delta_r\mu/\delta n > 0$, in reality the self-consistent mean field band structure changes with n in such a way that the net energy is minimised. The associated energy reduction is of second order in δn , which yields a negative term in the compressibility.

The EM phase is found in the region where the quasi-particle density of states at the Fermi level is strongly increased due to hybridisation, meaning $\xi_0 < 0$ for $n < 1$, $\xi_0 > 0$ for $n > 1$ (*i.e.*, the Fermi level lies between the crossing, $\xi(\epsilon) = 0$, of the Hartree $\Delta \rightarrow 0$ bands and the EM energy gap). In the $U\Delta \ll |\xi_0|$ limit, the term $\delta_r\mu/\delta n$ then vanishes, whereas Y_1 and hence $\delta_c\mu/\delta n$ remain finite. Similarly in the $U\Delta \gg 1$ limit, $\delta_r\mu/\delta n$ remains finite whereas $\delta_c\mu/\delta n \propto -U\Delta$. Both limits are formal since they can not arise as mean-field solutions; the real $\Delta \rightarrow 0$ regime at $t' = 0$ corresponds to either $n_d \rightarrow 0$ or $n_d \rightarrow 1$. Yet the fact that the negative $\delta_c\mu$ dominates in both cases, along with the results of our numerical investigation of actual mean field solutions for the 2D case, suggests that this is indeed the generic behaviour, both in $t' = 0$ and $t' \neq 0$ cases.

There is however one important exception, corresponding to the 3D case just above or just below half-filling. In the limit $n \rightarrow 1$, owing to vanishing $\nu(\epsilon)$ at the band edge, the dominant term is $\delta_r\mu \propto (\delta n)^{2/3}$. This results in a *positive* compressibility, presumably within an immediate vicinity of half-filling[40].

Based on the above discussion of EM compressibility in an oversimplified model, we can formulate tentative conclusions which perhaps are not restricted to the EFKM as described by Eq.(1): (i) $\delta\mu$ contains two terms, corresponding to filling a rigid hybridised bandstructure and to correlation effects. The correlations term is negative, always linear in δn (because it comes from a variation of Fermi-sea integrals) and typically dominant, resulting in a negative $\partial\mu/\partial n$ (ii) In three dimensions, whenever the Fermi level approaches a band edge or a Van Hove anomaly, the compressibility turns positive, presumably within a narrow range of doping values. Thus the EM *may* be thermodynamically stable in this region, provided that there are no other instabilities. In this regard, we note that the purported excitonic ferromagnetism in hexaborides is limited to small values of doping, although the difference in both physical systems and theoretical approaches does not allow for a direct analogy (see discussion in Sec. VI).

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